

Long Hoang Pham

VISION RESEARCHER · SURVEILLANCE SYSTEM ENGINEER

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“Simplicity is prerequisite for reliability.”

Summary

A computer vision researcher focused on robust scene understanding in adverse weather and edge AI technologies for intelligent surveillance systems. Studying deep learning algorithms robust to diverse lighting conditions, domain shifts, and occlusion, with the goal of achieving reliable real-world deployment on embedded hardware. An enthusiast of open-source research and reproducible engineering, aiming to contribute high-impact vision models to the broader AI community. Currently, a Research Professor at the [Automation Lab at SKKU](#), Republic of Korea.

Highlights

- Published across top-tier venues in computer vision (CVPR, ICCV, ECCV, AAAI, ACCV), intelligent surveillance systems (IEEE T-ITS, IEEE Access), and industrial systems (IEEE T-IIIE).
- Inventor on granted/pending patents and executed R&D deliverables for government agencies (IITP, NRF) and industry partners (Hyundai, Samsung).
- Academic honors & awards:
 - 2021 IEEE ICEIC Best Paper Award (among 300 papers, only 3 papers were selected)
 - 2017 IEEE ICSSE Best Paper Award (among 155 papers)
- Won multiple computer vision challenges and competitions while pursuing technologies for real-world applications:
 - 1st prize in [10th AI City Challenge](#), in ECCV'26, Sweden
 - 2nd prize in [9th AI City Challenge](#) in CVPR'25, USA
 - 1st prize in [1st TP-MOT Challenge](#) in ICCV'25, USA
 - 2nd prize in [8th AI City Challenge](#) in CVPR'24, USA
 - 1st prize in [7th AI City Challenge](#) in CVPR'23, Canada
 - 3rd prizes in [5th](#) and [6th UG2+ Challenges](#) in CVPR'22 & CVPR'23
 - Research Professor and Postdoctoral Fellow of Automation Lab, SKKU (advisor: [Prof. Jae Wook Jeon](#))

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Intelligent surveillance systems using camera and LiDAR sensors
- Multi-object detection and tracking with occlusion reasoning
- Low-light image enhancement and adverse weather perception
- Real-time scene understanding and semantic segmentation for autonomous systems
- Machine vision for automated industrial manufacturing and quality inspection
- Computer vision solutions for automated retail and self-checkout systems
- Edge-based real-time image and video processing

Education

Sungkyunkwan University (SKKU)

Suwon, Republic of Korea

PH.D. IN COMPUTER ENGINEERING

Aug. 2017 – Aug. 2021

- Dissertation: “A Smartphone-based Laser Measuring System for Gap and Flush Assessment in Car Body”
- Advised by [Prof. Jae Wook Jeon](#)

Vietnam National University HCMC –International University (HCMIU)

Vietnam

M.ENG. IN INFORMATION TECHNOLOGY MANAGEMENT

Nov. 2014 – Aug. 2017

- University Valedictorian (Ranked 1st across all Master's programs, GPA: 3.8/4.0)
- Dissertation: “A Robust Multiclass Vehicle Detection and Classification Algorithm for Traffic Surveillance System”
- Advised by [Dr. Synh Viet-Uyen Ha](#)

Vietnam National University HCMC –International University (HCMIU)

Vietnam

B.S. IN COMPUTER SCIENCE

Aug. 2009 – May 2014

- Department Salutatorian (Ranked 2nd in Computer Science, GPA: 3.85 / 4.0)
- Dissertation: “Extended Segmentation Algorithm in Traffic Monitoring System”
- Advised by [Dr. Synh Viet-Uyen Ha](#)

Work Experience

Research Professor & Postdoctoral Fellow

AUTOMATION LAB, SKKU

Suwon, Republic of Korea

Sep. 2021 – Current

- Advised by [Prof. Jae Wook Jeon](#).
- Prepared and was granted 2 Korean government research grants.
 - “An Intelligent System for 24/7 Real-Time Traffic Surveillance on Edge Devices” (8 years).
 - “Real-Time Scene Understanding Technology in Complex Traffic Conditions and in Bad Weather” (3 years).
- Researching deep learning models for robust outdoor, real-world surveillance tasks.
- Developing robust image processing algorithms for industrial manufacturing and quality control.
- Led a team to attend and win several international challenges at top conferences (CVPR, ICCV, ECCV, AAAI, ACCV).
- Guiding and supervising graduate and undergraduate students in their research papers and thesis projects.

Research Assistant

COMPUTER SCIENCE & ENGINEERING, HCMIU

Vietnam

Aug. 2015 – Aug. 2017

- Joined a city-funded research project “Piloting Project on the Intelligent Traffic System of Ho Chi Minh City”.
- Developed and deployed a real-time surveillance system to extract traffic flow using CCTVs.
- Guided and supervised undergraduate students in their research papers and thesis projects.

Research Project Experience

An Intelligent System for 24/7 Real-Time Traffic Surveillance on Edge Devices

[Automation Lab, SKKU](#)

SUPPORTED BY IITP, WHICH IS A KOREAN GOVERNMENT-AFFILIATED ORGANIZATION

2021 – Current

- Researched robust computer vision algorithms for severe environmental domain shifts (nighttime, rain, fog, and glare)
- Developed lightweight deep learning models to achieve low-latency processing on embedded hardware
- Built a large-scale annotated traffic surveillance dataset

Real-Time Scene Understanding Technology in Complex Traffic and Bad Weather

[Automation Lab, SKKU](#)

SUPPORTED BY NATIONAL RESEARCH FOUNDATION, REPUBLIC OF KOREA

2020 – 2022

- Developed semantic segmentation models for real-time scene parsing under adverse weather (rain, fog, snow, and low-light)
- Formulated feature enhancement to counteract image degradation and maintain high pixel-level classification accuracy
- Optimized deep learning models for real-time processing on embedded autonomous hardware

Development of A Smartphone-based Laser Measurement System for Assessment of Gap, Flush, and Curvature in Car Body

[Automation Lab, SKKU](#)

SUPPORTED BY HYUNDAI MOTOR COMPANY CO., LTD.

2018 – 2021

- Engineered a handheld laser profiling device leveraging a smartphone as both the primary camera sensor and real-time processing unit
- Developed computer vision algorithms for sub-pixel laser stripe extraction, displacement calibration, and surface curvature estimation
- Formulated camera-laser triangulation models to accurately measure gap and flush tolerances across car body panels
- Collaborated closely with industry partners to evaluate system accuracy against industrial-grade measurement tools

Advanced Occlusion Detection Algorithm in Video Processing Applications

[CVIP Lab, HCMIU](#)

SUPPORTED BY VIETNAM NATIONAL UNIVERSITY - HCM

2016 – 2017

- Developed computer vision algorithms for real-time vehicle occlusion detection in traffic surveillance videos
- Implemented spatiotemporal feature analysis to distinguish between temporary object overlap and completed target disappearance
- Enhanced multi-object tracking stability under partial and heavy occlusion scenarios across crowded video sequences

Extended Occlusion Vehicle Detection Algorithm in Crowded Traffic Scenes

[CVIP Lab, HCMIU](#)

STUDENT RESEARCH GRANT SUPPORTED BY HCMIU

2015 – 2016

- Investigated image processing algorithms to improve vehicle detection under severe visual occlusion
- Formulated spatial reasoning techniques to separate overlapping vehicle bounding boxes in congested surveillance video
- Supervised undergraduate students in experiment design and performance benchmarking

Piloting Project on the Intelligent Traffic System of Ho Chi Minh City

[CVIP Lab, HCMIU](#)

SUPPORTED BY HO CHI MINH CITY PEOPLE'S COMMITTEE

2015 – 2017

- Developed a city-wise traffic flow estimation system for vehicle counting and speed estimation
- Optimized software infrastructure to ensure low-latency inference and seamless data integration across a citywide CCTV network
- Implemented core computer vision algorithms and backend processing infrastructure
- Media coverage: [DanTri \(Aug. 25, 2017\)](#)

Video Processing Technologies for Intelligent Traffic Monitoring Systems

[CVIP Lab, HCMIU](#)

RESEARCH GRANT SUPPORTED BY HCMIU

2014 – 2015

- Implemented background subtraction, motion estimation, and blob analysis pipelines for robust vehicle detection
- Developed video analytics modules to extract traffic flow in CCTV videos
- Contributed as a core developer to an institutional research grant funded by HCMIU

- Investigated image segmentation algorithms for traffic surveillance tasks
- Combined optical flow with an automatic observation zone detection algorithm and ellipse feature fitting to classify vehicles
- Developed a real-time image processing system to count moving vehicles in pre-recorded traffic surveillance video

Publications

PATENTS

- [1] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, *AUTOMATIC CHECKOUT SYSTEM AND CONTROL METHOD FOR THE SAME*. 자동계산대시스템및그제어방법. Patent 10-2994156. Sungkyunkwan University Research & Business Foundation, Application No. 10-2022-0154620, Publication No. 10-2024-0072708, Filed Nov 17, 2022, Published May 24, 2024, Registered Jul. 16, 2026. July 2026. doi: [10.8080/1020220154620](https://doi.org/10.8080/1020220154620).
- [2] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, Huy-Hung Nguyen, *DEVICE AND METHOD FOR DETECTING ON PRODUCT IN AUTOMATIC CHECKOUT SYSTEM, STORAGE MEDIUM STORING INSTRUCTIONS TO PERFORM METHOD FOR DETECTING ON PRODUCT IN AUTOMATIC CHECKOUT SYSTEM*. Patent US-12682334-B2. Sungkyunkwan University Research & Business Foundation, Application No. 18/913,280, Filed Oct. 11, 2024, Registered Jul. 14, 2026. July 2026.
- [3] Jae Wook Jeon, Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Huy-Hung Nguyen, Tai Huu-Phuong Tran, Hyung-Min Jeon, *APPARATUS FOR JUDGING VIOLATION OF TWO-WHEELED VEHICLE EXCLUSIVE LANE AND ITS OPERATION METHOD*. 이륜차전용차로위반여부판단장치및그동작방법. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 10-2024-0179946, Publication No. 10-2026-0088248, Filed Dec. 05, 2024, Published Jun. 12, 2026. June 2026. doi: [10.8080/1020240179946](https://doi.org/10.8080/1020240179946).
- [4] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, *METHOD FOR PREDICTING TRAFFIC INFORMATION AND APPARATUS THEREOF*. 교통정보예측방법및장치. Patent 10-2833871. Sungkyunkwan University Research & Business Foundation, Application No. 10-2022-0047198, Publication No. 10-2023-0148050, Filed Apr. 15, 2022, Published Oct. 24, 2023, Registered Jul. 09, 2025. July 2025. doi: [10.8080/1020220047198](https://doi.org/10.8080/1020220047198).
- [5] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, Huy-Hung Nguyen, *METHOD AND DEVICE FOR PRODUCT DETECTING OF AUTOMATIC CHECKOUT SYSTEM*. 자동계산대시스템의상품검출방법및이를위한상품검출장치. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 10-2023-0135771, Publication No. 10-2025-0052688, Filed Oct. 12, 2023, Published Apr. 21, 2025. Apr. 2025. doi: [10.8080/1020230135771](https://doi.org/10.8080/1020230135771).
- [6] Jae Wook Jeon, **Long Hoang Pham**, Huy-Hung Nguyen, Duong Nguyen-Ngoc Tran, Hyung-Min Jeon, *GUIDED CURVE ESTIMATION METHOD AND APPARATUS FOR ENHANCING LOW-LIGHT IMAGES*. 저조명도영상개선을위한가이드커브추정방법및장치. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 10-2024-0073935, Publication No. 10-2025-0174395, Filed Jun. 05, 2024, Published Dec. 12, 2025. Dec. 2025. doi: [10.8080/1020240073935](https://doi.org/10.8080/1020240073935).
- [7] Jae Wook Jeon, Duong Nguyen-Ngoc Tran, Hyung-Min Jeon, **Long Hoang Pham**, Huy-Hung Nguyen, *CENTRAL DATA PROCESSING APPARATUS USING PLURALITY OF DATA PROCESSING APPARATUSES AND DATA PROCESSING METHOD USING THE SAME*. 다수의데이터처리장치를이용하는중앙데이터처리장치및이를이용한데이터처리방법. Patent 10-2824453. Sungkyunkwan University Research & Business Foundation, Application No. 10-2022-0076667, Publication No. 10-2024-0000117, Filed Jun. 23, 2022, Published Jan. 02, 2024, Registered Jun. 19, 2025. June 2025. doi: [10.8080/1020220076667](https://doi.org/10.8080/1020220076667).
- [8] Jae Wook Jeon, Duong Nguyen-Ngoc Tran, Hyung-Min Jeon, **Long Hoang Pham**, Huy-Hung Nguyen, *METHOD AND APPARATUS FOR PROVIDING OBJECT RECOGNITION, LEARNING METHOD FOR OBJECT RECOGNITION*. 객체인식장치및인식방법, 객체인식을위한학습방법. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 10-2023-0132012, Publication No. 10-2025-0049099, Filed Oct. 10, 2023, Published Apr. 11, 2025. Apr. 2025. doi: [10.8080/1020230132012](https://doi.org/10.8080/1020230132012).
- [9] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Huy-Hung Nguyen, Duong Nguyen-Ngoc Tran, *Method and device for enhancing low-light images*. 저조명도영상을개선하는방법및장치. Patent 10-2695217. Sungkyunkwan University Research & Business Foundation, Application No. 10-2023-0051853, Filed Apr. 20, 2023, Registered Aug. 09, 2024. Aug. 2024. doi: [10.8080/1020230051853](https://doi.org/10.8080/1020230051853).
- [10] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, *AUTOMATIC CHECKOUT SYSTEM, METHOD FOR CONTROLLING AUTOMATIC CHECKOUT SYSTEM, AND STORAGE MEDIUM STORING INSTRUCTIONS TO PERFORM METHOD FOR CONTROLLING AUTOMATIC CHECKOUT SYSTEM*. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 18/512,206, Publication No. US-20240312185-A1, Filed Nov. 17, 2023, Published Sep. 19, 2024. Sept. 2024.
- [11] Jae Wook Jeon, Duong Nguyen-Ngoc Tran, Hyung-Min Jeon, **Long Hoang Pham**, Huy-Hung Nguyen, *Apparatus and method for generating data for learning of artificial neural network*. 인공신경망의학습을위한데이터를생성하는장치및방법. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 10-2023-0006155, Publication No. 10-2024-0114129, Filed Jan. 16, 2023, Published Jul. 23, 2024. July 2024. doi: [10.8080/1020230006155](https://doi.org/10.8080/1020230006155).
- [12] Jae Wook Jeon, Duong Nguyen-Ngoc Tran, Hyung-Min Jeon, **Long Hoang Pham**, Huy-Hung Nguyen, *APPARATUS AND METHOD FOR GENERATING DATA FOR TRAINING OF NEURAL NETWORK AND STORAGE MEDIUM STORING INSTRUCTIONS TO PERFORM METHOD FOR GENERATING DATA FOR TRAINING OF NEURAL NETWORK*. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 18/413,222, Publication No. US-20240428574-A1, Filed Jan. 16, 2024, Published Dec. 26, 2024. Dec. 2024.
- [13] Jae Wook Jeon, **Long Hoang Pham**, Hyung-Min Jeon, Duong Nguyen-Ngoc Tran, *METHOD FOR PREDICTING TRAFFIC INFORMATION, APPARATUS FOR PREDICTING TRAFFIC INFORMATION, AND STORAGE MEDIUM STORING INSTRUCTIONS TO PERFORM METHOD FOR PREDICTING TRAFFIC INFORMATION*. Patent Application. Sungkyunkwan University Research & Business Foundation, Application No. 18/133,630, Publication No. US-20230334980-A1, Filed Apr. 12, 2023, Published Oct. 19, 2023. Oct. 2023.

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- [1] Huy-Hung Nguyen, Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Jae Wook Jeon, *Optimizing Monocular Driving Assistance for Real-Time Processing on Jetson AGX Xavier*. In: *IEEE Access* 12 (2024), pp. 71853–71865. ISSN: 2169-3536. doi: [10.1109/ACCESS.2024.3402239](https://doi.org/10.1109/ACCESS.2024.3402239).
- [2] Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Huy-Hung Nguyen, Jae Wook Jeon, *A Vision-Based Method for Real-Time Traffic Flow Estimation on Edge Devices*. In: *IEEE Transactions on Intelligent Transportation Systems* 24.8 (2023), pp. 8038–8052. ISSN: 1524-9050. doi: [10.1109/TITS.2023.3264796](https://doi.org/10.1109/TITS.2023.3264796).
- [3] Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Huy-Hung Nguyen, Tai Huu-Phuong Tran, Hyung-Joon Jeon, Jae Wook Jeon, *Universal Detection-Based Driving Assistance Using a Mono Camera With Jetson Devices*. In: *IEEE Access* 10 (2022), pp. 59400–59412. ISSN: 2169-3536. doi: [10.1109/ACCESS.2022.3179999](https://doi.org/10.1109/ACCESS.2022.3179999).
- [4] **Long Hoang Pham**, Duong Nguyen-Ngoc Tran, Jin Young Byun, Chul Hong Rhie, Jae Wook Jeon, *A Smartphone-Based Laser Measuring System for Gap and Flush Assessment in Car Body*. In: *IEEE Transactions on Industrial Electronics* 68.7 (July 2021), pp. 6297–6307. doi: [10.1109/TIE.2020.2992971](https://doi.org/10.1109/TIE.2020.2992971).
- [5] Synh Viet-Uyen Ha, **Long Hoang Pham**, Ha Manh Tran, Phong Ho, *Robust Reflection Detection and Removal in Rainy Conditions using LAB and HSV Color Spaces*. In: *REV Journal on Electronics and Communications* 6.1–2 (2016). ISSN: 1859-378X. doi: [10.21553/rev-jec.130](https://doi.org/10.21553/rev-jec.130).
- [6] Synh Viet-Uyen Ha, **Long Hoang Pham**, Ha Manh Tran, Phong Ho, *Improved Vehicles Detection & Classification Algorithm for Traffic Surveillance System*. In: *Journal of Information Assurance & Security* 9.7 (2014), pp. 385–392.

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- [1] Ngoc Doan-Minh Huynh, Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Tai Huu-Phuong Tran, Hyung-Joon Jeon, Huy-Hung Nguyen, Duong Khac Vu, Hyung-Min Jeon, Son Hong Phan, Quoc Pham-Nam Ho, Chi Dai Tran, Trinh Le Ba Khanh, Jae Wook Jeon, “TSBOW: Traffic Surveillance Benchmark for Occluded Vehicles Under Various Weather Conditions”. In: *AAAI Conference on Artificial Intelligence*. Singapore, Mar. 2026, pp. 5239–5247. doi: [10.1609/aaai.v40i7.37439](https://doi.org/10.1609/aaai.v40i7.37439) .
- [2] **Long Hoang Pham**, Quoc Pham-Nam Ho, Huy-Hung Nguyen, Duong Nguyen-Ngoc Tran, Ngoc Doan-Minh Huynh, Cu Quoc Le, Hoang-Khang Nguyen, Hyung-Min Jeon, Chi Dai Tran, Son Hong Phan, Duong Khac Vu, Trinh Le Ba Khanh, Jae Wook Jeon, “Rethinking Pre-Training and Augmentation for Zero-Shot Cross-City Object Detection”. In: *European Conference on Computer Vision Workshop (ECCVW)*. Malmö, Sweden, Sept. 2026 .
- [3] Duong Nguyen-Ngoc Tran, Ngoc Doan-Minh Huynh, Cu Quoc Le, Hoang-Khang Nguyen, **Long Hoang Pham**, Huy-Hung Nguyen, Quoc Pham-Nam Ho, Trinh Le Ba Khanh, Chi Dai Tran, Duong Khac Vu, Son Hong Phan, Hyung-Min Jeon, Jae Wook Jeon, “Syn2RealTrack: Bridging the Gap Between Synthetic and Real World Dataset for Online Multi-View Multi-Target Tracking”. In: *European Conference on Computer Vision Workshop (ECCVW)*. Malmö, Sweden, Sept. 2026 .
- [4] Duong Nguyen-Ngoc Tran, **Long Hoang Pham**, Quoc Pham-Nam Ho, Chi Dai Tran, Ngoc Doan-Minh Huynh, Huy-Hung Nguyen, Jae Wook Jeon, “A Hybrid Data-Centric Framework for Thermal Multiple-Object Tracking with Complex Motion Patterns”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. Denver, Colorado, USA: IEEE, June 2026, pp. 7200–7209 .
- [5] **Long Hoang Pham**, Quoc Pham-Nam Ho, Duong Khac Vu, Huy-Hung Nguyen, “Data Augmentation Is All You Need For Robust Fisheye Object Detection”. In: *IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*. Honolulu, HI, USA: IEEE, Oct. 2025, pp. 5393–5401. ISBN: 979-8-3315-8988-2. doi: [10.1109/ICCVW69036.2025.00563](https://doi.org/10.1109/ICCVW69036.2025.00563) .
- [6] Tai Huu-Phuong Tran, Duong Nguyen-Ngoc Tran, Ngoc Doan-Minh Huynh, Chi Dai Tran, **Long Hoang Pham**, Quoc Pham-Nam Ho, Huy-Hung Nguyen, Duong Khac Vu, Hyung-Min Jeon, Hyung-Joon Jeon, Son Hong Phan, Trinh Le Ba Khanh, Jae Wook Jeon, “DepthTrack: Cluster Meets BEV for Multi-Camera Multi-Target 3D Tracking”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. Honolulu, HI, USA: IEEE, Oct. 2025, pp. 5348–5357. ISBN: 979-8-3315-8988-2. doi: [10.1109/ICCVW69036.2025.00558](https://doi.org/10.1109/ICCVW69036.2025.00558) .
- [7] Trinh Le Ba Khanh, Huy-Hung Nguyen, **Long Hoang Pham**, Duong Nguyen-Ngoc Tran, Jae Wook Jeon, “Dynamic Retraining-Updating Mean Teacher for Source-Free Object Detection”. In: *Computer Vision – ECCV 2024*. Ed. by Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, Gül Varol, vol. 15110. Milan, Italy: Springer Nature Switzerland, Sept. 2024, pp. 328–344. ISBN: 978-3-031-72943-0. doi: [10.1007/978-3-031-72943-0_19](https://doi.org/10.1007/978-3-031-72943-0_19) .
- [8] Huy-Hung Nguyen, Chi Dai Tran, **Long Hoang Pham**, Duong Nguyen-Ngoc Tran, Tai Huu-Phuong Tran, Duong Khac Vu, Quoc Pham-Nam Ho, Ngoc Doan-Minh Huynh, Hyung-Min Jeon, Hyung-Joon Jeon, Jae Wook Jeon, “Multi-View Spatial-Temporal Learning for Understanding Unusual Behaviors in Untrimmed Naturalistic Driving Videos”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. Seattle, WA, USA: IEEE, June 2024, pp. 7144–7152. ISBN: 979-8-3503-6547-4. doi: [10.1109/CVPRW63382.2024.00709](https://doi.org/10.1109/CVPRW63382.2024.00709) .
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OTHERS

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Teaching Experience

[ECE5759] Intelligent Control (Lecturer: Prof. Jae Wook Jeon)

Electrical and Computer
Engineering, SKKU
Spring, 2022 – 2023

TEACHING ASSISTANT

- Graded assignments, tests, and final project for graduate students.
- Guided and interacted with students during a 2-week final project.

Competitive Programming (Lecturer: Dr. Synh Viet-Uyen Ha)

Computer Science & Engineering,
HCMIU
Aug. 2015 – Aug. 2017

INSTRUCTOR

- Shared teaching responsibilities with other instructors; together we trained 20 undergraduate students.
- Led teams to attend: [Vietnam Student Olympiad in Informatics \(VNOI\)](#) and [ACM-ICPC](#).

[IEIEIU24033] Introduction to Industrial Systems

Industrial and Systems Engineering,
HCMIU
Spring, 2015

TEACHING ASSISTANT

- Taught weekly lab sessions on C/C++ programming for undergraduate students.
- Provided hands-on support for students to perform a term-long project.

Honors & Awards

INTERNATIONAL AWARDS

2026	1st Prize (Cross-City Object Detection), 10th AICity , in ECCV'26	Malmö, Sweden
2026	2nd Prize (Multi-Camera 3D Perception), 10th AICity , in ECCV'26	Malmö, Sweden
2025	2nd Prize (Road Object Detection in Fish-Eye Cameras), 9th AICity , in ICCV'25	Honolulu, HI, USA
2025	2nd Prize (Multi-Camera 3D Perception), 9th AICity , in ICCV'25	Honolulu, HI, USA
2025	1st Prize, 1st TP-MOT, 21st PBVS Workshop in CVPR'25	Nashville, TN, USA
2024	3rd Prize (Road Object Detection in Fish-Eye Cameras), 8th AICity , in CVPR'24	Seattle, WA, USA
2024	2nd Prize (Naturalistic Driving Action Recognition), 8th AICity , in CVPR'24	Seattle, WA, USA
2023	1st Prize (Detecting Violation of Helmet Rule for Motorcyclists), 7th AICity , in CVPR'23	Vancouver, BC, Canada
2023	3rd Prize (Object Detection in Haze), 6th UG2+ , in CVPR'23	Vancouver, BC, Canada
2022	3rd Prize (Object Detection in Haze), 5th UG2+ , in CVPR'22	New Orleans, LA, USA
2021	Best Paper Award Silver Prize (among 300 papers, 3 papers were received) , IEEE ICEIC'21	Jeju, Republic of Korea
2017	Best Paper Award (among 155 papers) , IEEE ICSSE'17	Ho Chi Minh City, Vietnam

DOMESTIC AWARDS (REPUBLIC OF KOREA)

2023	Top 100 Outstanding National R&D Achievements , Korean Minister of Science and ICT	Seoul
2020	1st Prize (Instance Segmentation) , Driving Image AI Learning Dataset Contest , KATECH	Chungcheongnam-do
2020	2nd Prize (Object Detection) , Driving Image AI Learning Dataset Contest , KATECH	Chungcheongnam-do
2020	1st Prize (Forward 2D) , Automated Driving Recognition Technology Competition , in 4th Pangyo Autonomous Mobility Show	Gyeonggi-do
2020	1st Prize (Omni-directional 3D) , Automated Driving Recognition Technology Competition , in 4th Pangyo Autonomous Mobility Show	Gyeonggi-do
2019	2nd Prize (Forward 2D & Omni-directional 3D) , Automated Driving Recognition Technology Competition, in 3rd Pangyo Autonomous Mobility Show	Gyeonggi-do
2017	Full Scholarship , Ph.D. in Computer Engineering, SKKU	Suwon

DOMESTIC AWARDS (VIETNAM)

2017	Valedictorian , Graduate Program (Intake 2015–2017) , HCMIU	Ho Chi Minh City
2017	Excellent Scientific Publications Award , Graduate Program (Intake 2015–2017) , HCMIU	Ho Chi Minh City
2015	Toshiba Scholarship for Graduate Students , HCMIU	Ho Chi Minh City
2014	Full Scholarship , M.Eng. in Information Technology Management, HCMIU	Ho Chi Minh City
2017	Salutatorian , CS (Class 2009–2014), HCMIU	Ho Chi Minh City
2014	Ho Chi Minh City ICT Award , HCMIU	Ho Chi Minh City
2013	Rector's Award for Academic Excellence , HCMIU	Ho Chi Minh City
2013	3rd Prize , Vietnam Student Olympiad in Informatics	Da Nang
2013	Honorable Mention , ACM/ICPC Asia Regional Contest 2013	Da Nang
2013	Pony Chung Scholarship , HCMIU	Ho Chi Minh City
2012	Consolation Prize , Vietnam Student Olympiad in Informatics	Hanoi
2012	Honorable Mention , ACM/ICPC Asia Regional Contest 2012	Hanoi
2012	CSC Vietnam Scholarship , HCMIU	Ho Chi Minh City
2009	University Entrance Scholarship , HCMIU	Ho Chi Minh City

Academic Services

2027	Reviewer , WACV
2026	Reviewer , Pattern Recognit., PIIIEE, MDPI Sensors, CVPR, ECCV, BMVC, AAAI, ACM-MM, WACV, MAPR
2025	Reviewer , PIIIEE, MDPI Electronics, CVPR, ICCV, AAAI, ACM-MM, NeurIPS
2024	Reviewer , IEEE T-ITS, MDPI Electronics, MDPI Sensors, CVPR, ECCV, ACCV, NeurIPS
2023	Reviewer , IEEE T-ITS, IEEE Access, CVPR, ICCV
2022	Reviewer , IEEE T-ITS, IEEE Access
2020	Reviewer , IEEE TIE, IET-ITS
2019	Reviewer , IET-ITS

Skills

Programming	C/C++, Python, Swift, Objective-C, LaTeX, R
Frameworks	OpenCV, PyTorch, TensorRT, Git, Docker
Database	MySQL, Microsoft SQL
Front-end	HTML5, Qt
Languages	Vietnamese, English